

Low Level Design (LLD)

1. Component Module Mapping

The system comprises modular, cohesive python modules:

Component Class	File Path	Responsibilities
DatabaseManager	app/database/database.py	SQLite schemas, column migrations, users management
ModelInference	app/services/predict.py	Load preprocessors, model scoring execution
LIMEService	app/services/predict.py	Ridge local surrogate explainability calculations